Status	Finished
Started	Thursday, 8 October 2026, 9:48 AM
Completed	Thursday, 8 October 2026, 9:53 AM
Duration	5 mins 1 sec
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Quick Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Duplicates
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **3**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is difficult to implement
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Finding a specific element in a list
- c. Arranging data in a specific order
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It allows for quicker identification of duplicates
- c. It has no effect on the analysis process
- d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Backtracking
- c. Greedy method
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Selection
- c. Distribution
- d. Extraction

The correct answer is:
Insertion

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Sorting

c. Complexity

d. Merging

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort

b. Heap sort

c. Merge sort

d. Selection sort

The correct answer is:
Merge sort

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are compared element by element
- b. They are discarded
- c. They are combined to form a single sorted list
- d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It increases the number of comparisons needed
- c. It makes the search operation slower
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **14**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sorted()

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It builds a sorted array one element at a time
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them